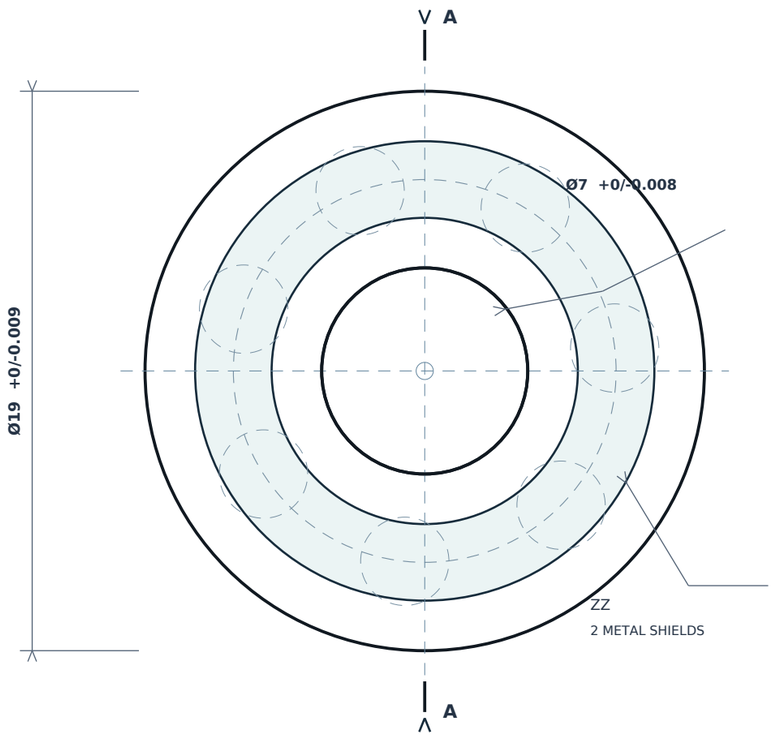
MECHANICAL ASSEMBLY DRAWING / TRAINING SAMPLE

607ZZ SHIELDED MINIATURE BEARING

NOT FOR MANUFACTURE / NOT APPROVED

SOURCE-BASED ENVELOPE / EXAMPLE INTERNAL DESIGN

01 FRONT ELEVATION

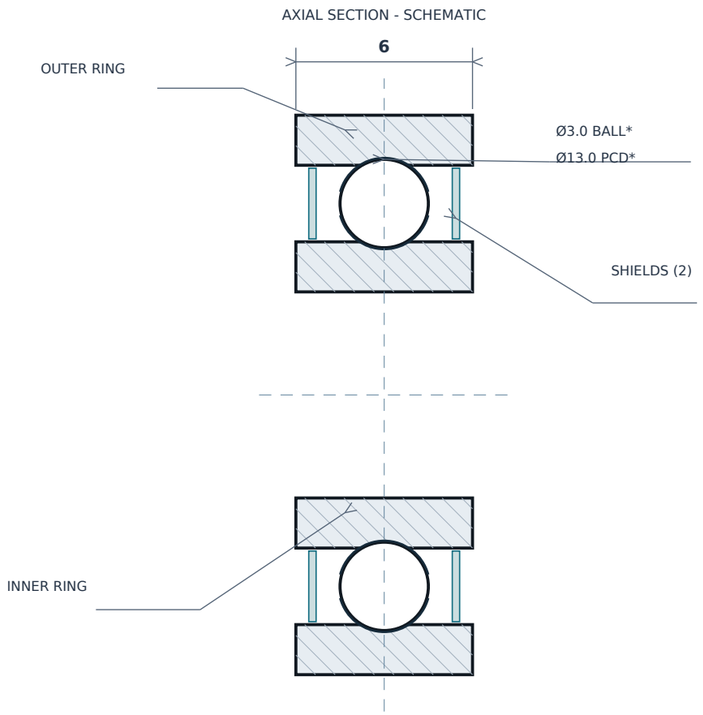


VIEW 01 | FACE (4.8:1)

02 AXIAL PROFILE

VIEW 02 | PROFILE (4.8:1)

03 SECTION A-A



VIEW 03 | CUT A-A (4.8:1)

SOURCE-CONTROLLED FEATURES / SELECTED P0 CLASS

BORE	OD	WIDTH	EDGE DETAIL	GRADE
Ø7 +0/-0.008	**Ø19 +0/-0.009**	**6 +0/-0.120**	**0.3 mm MIN**	**P0 (EXAMPLE)**

Supplier P0 values; tolerance grade and radial play class must be confirmed for the actual order.

CONSTRUCTION

Rings and balls: SAE 52100 chrome steel.
Cage: pressed steel. Closures: two steel shields.
Standard grease: Kyodo Yushi Multemp SRL**.

PERFORMANCE / SOURCE

Static 87 kgf; dynamic 228 kgf; max 36,000 rpm*.
Radial play: selectable MC1-MC6 (not specified).
*With adequate lubrication. **Grease may vary.

STATUS

Reference-only example. Supplier dimensions shown separately from assumed demonstration geometry.

All numerical internal details marked * are illustrative. Obtain approved component drawings before production.

PROJECTION

Third-angle orthographic (illustrated). Dimensions govern; do not scale print.

DOCUMENT CONTROL Educational CAD sample / no engineering sign-off

DRAWING NUMBER
607ZZ-DEMO-1

PART / DESCRIPTION	REV / STATUS	SCALE
607ZZ - RADIAL BALL BEARING	**A / SAMPLE**	**AS NOTED**
UNITS	SHEET	SOURCE
mm	**1 OF 2**	**SMB / 607ZZ**

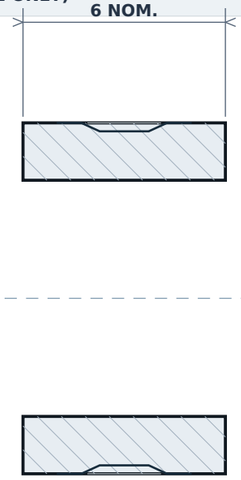
MECHANICAL ASSEMBLY DRAWING / TRAINING SAMPLE

NOT FOR MANUFACTURE / NOT APPROVED

ILLUSTRATIVE COMPONENTS & INSPECTION SCHEDULE

ASSUMPTIONS ARE NOT SUPPLIER-APPROVED MANUFACTURING DATA

DETAIL B / INNER RING (NTS / EXAMPLE ONLY)

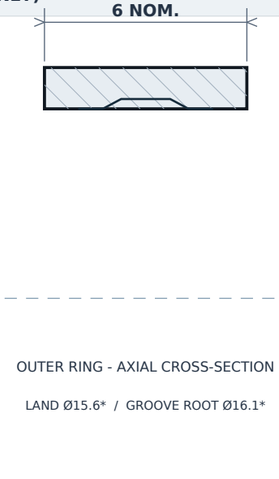


INNER RING - AXIAL CROSS-SECTION

LAND Ø10.4* / GROOVE ROOT Ø9.9*

BORE Ø7, +0/-0.008 [SOURCE]

DETAIL C / OUTER RING (NTS / EXAMPLE ONLY)



OUTER RING - AXIAL CROSS-SECTION

LAND Ø15.6* / GROOVE ROOT Ø16.1*

OD Ø19, +0/-0.009 [SOURCE]

ASSEMBLY PARTS SCHEDULE

ITEM	COMPONENT	QTY	REFERENCE MATERIAL
01	Outer race	1	SAE 52100
02	Inner race	1	SAE 52100
03	Steel balls*	7	SAE 52100
04	Retainer / cage	1	Pressed steel
05	Metal shields	2	Metal shield
06	Grease	AR	Multemp SRL**

INSPECTION PLAN / P0 SOURCE VALUES

CHARACTERISTIC	ALLOWABLE	METHOD (SAMPLE)
Bore Ø7	6.992 to 7.000 mm	Bore comparator
OD Ø19	18.991 to 19.000 mm	OD micrometer
Total width 6	5.880 to 6.000 mm	Micrometer
Inner radial runout	0.010 mm max	Dial test indicator
Outer radial runout	0.015 mm max	Dial test indicator
Clearance class	MC1-MC6; select	Class-specific test

EXAMPLE DESIGN ASSUMPTIONS* (NOT IN THE SUPPLIER PDF)

7 balls x Ø3.0 mm on Ø13.0 mm PCD; inner shoulder Ø10.4 / groove root Ø9.9; outer land Ø15.6 / groove root Ø16.1.

Groove shape, cage stamping, shield locking and grease fill are schematic; treat all as unvalidated engineering assumptions.

Additional release data required: raceway radii/profile, concentricity, hardness/heat treatment, roughness, ball grade, shield fit, clearances and quality plan.

STATUS

Reference-only example. Supplier dimensions shown separately from assumed demonstration geometry.

All numerical internal details marked * are illustrative. Obtain approved component drawings before production.

PROJECTION

Third-angle orthographic (illustrated). Dimensions govern; do not scale print.

DOCUMENT CONTROL Educational CAD sample / no engineering sign-off

DRAWING NUMBER

607ZZ-DEMO-2

PART / DESCRIPTION	REV / STATUS	SCALE
607ZZ - RADIAL BALL BEARING	**A / SAMPLE**	**AS NOTED**
UNITS	SHEET	SOURCE
mm	**2 OF 2**	**SMB / 607ZZ**